SOC FINAL INCIDENT REPORT

Lab 3: Suspicious Office Document & LOLBin Execution

1. Incident Overview & Triage Metadata

Incident Title	Macro-Enabled Excel Phishing Attachment & LOLBin Execution (certutil.exe / rundll32.exe)
Assigned Severity	**CRITICAL**
Affected Host	FIN-WKSTN-04
Compromised Account	FIN-WKSTN-04\a.sharma
Incident Timeframe	2023-11-16 10:14:02 to 2023-11-16 10:14:33 (+05:30)
Detection Source	Splunk Enterprise (index="soc-investigation", source="*office_doc*")

2. Executive Summary

On November 16, 2023, at 10:14:02, user FIN-WKSTN-04\a.sharma received an email attachment in Microsoft Outlook (OUTLOOK.EXE, PID 2100) that was cached locally at C:\Users\a.sharma\AppData\Local\Microsoft\Windows\INetCache\Content.Outlook\Q3_Tax_Adjustment.xlsm. At 10:14:10, the user opened the macro-enabled workbook using Microsoft Excel (EXCEL.EXE, PID 4800).

At 10:14:18, a malicious macro inside the spreadsheet spawned a Windows Command Shell (cmd.exe, PID 5200), which chained two Living-off-the-Land Binaries (certutil.exe and rundll32.exe). certutil.exe (PID 5650) connected to http://203.0.113.88:8080/payload.dll and saved the remote file locally as C:\Users\a.sharma\AppData\Local\Temp\calc_helper.dll. Subsequently, rundll32.exe (PID 6100) executed the DLL payload, established an outbound C2 session to 203.0.113.88:443, executed local and domain discovery commands (whoami.exe and net.exe), and established user-level persistence by dropping OneDriveUpdater.lnk into the Windows Startup directory.

3. Reconstructed Process Ancestry & Attack Timeline

- OUTLOOK.EXE (PID 2100) — Cached Q3_Tax_Adjustment.xlsm (10:14:02 | EventCode 11)
- EXCEL.EXE (PID 4800) — Opened Q3_Tax_Adjustment.xlsm (10:14:10 | EventCode 4688)
- cmd.exe (PID 5200) — Macro shell execution (10:14:18 | EventCode 4688)
- certutil.exe (PID 5650) — Downloaded payload.dll (10:14:19 | EventCode 4688)
- Connected to 203.0.113.88:8080 (10:14:21 | EventCode 3)
- Dropped C:\Users\a.sharma\AppData\Local\Temp\calc_helper.dll (10:14:22 | EventCode 11)
- rundll32.exe (PID 6100) — Executed calc_helper.dll,StartWork (10:14:24 | EventCode 4688)
- Connected to C2 203.0.113.88:443 (10:14:26 | EventCode 3)
- Spawned whoami.exe /all (PID 6210 | 10:14:28 | EventCode 4688)
- Spawned net.exe user /domain (PID 6240 | 10:14:30 | EventCode 4688)
- Created Startup\OneDriveUpdater.lnk (10:14:33 | EventCode 11)

4. Confirmed MITRE ATT&CK Mapping

Timestamp	EventCode	Tactic	Technique ID & Name	Supporting Log Evidence
10:14:02	11	Initial Access	T1566.001 – Spearphishing Attachment	OUTLOOK.EXE (PID 2100) saved Q3_Tax_Adjustment.xlsm to the Outlook cache folder.
10:14:10	4688	Execution	T1204.002 – User Execution: Malicious File	EXCEL.EXE (PID 4800) launched Q3_Tax_Adjustment.xlsm from OUTLOOK.EXE.
10:14:18	4688	Execution	T1059.003 – Windows Command Shell	EXCEL.EXE (PID 4800) spawned cmd.exe /c ... (PID 5200).
10:14:19–22	4688, 3, 11	Command & Control	T1105 – Ingress Tool Transfer	certutil.exe (PID 5650) fetched http://203.0.113.88:8080/payload.dll and wrote calc_helper.dll to disk.
10:14:24	4688	Defense Evasion	T1218.011 – System Binary Proxy Execution: Rundll32	cmd.exe spawned rundll32.exe C:\Users\a.sharma\AppData\Local\Temp\calc_helper.dll,StartWork (PID 6100).
10:14:28	4688	Discovery	T1033 – System Owner/User Discovery	rundll32.exe spawned whoami.exe (PID 6210).
10:14:30	4688	Discovery	T1087.002 – Account Discovery: Domain Account	rundll32.exe spawned net.exe (PID 6240) to enumerate domain accounts.
10:14:33	11	Persistence	T1547.001 – Registry Run Keys / Startup Folder	rundll32.exe (PID 6100) dropped OneDriveUpdater.lnk in the Windows Startup folder.

5. Indicators of Compromise (IOCs)

Indicator Type	Value	Context
External IPv4 / Ports	203.0.113.88 (TCP 8080, TCP 443)	Staging server (8080) and outbound C2 channel (443)
Malicious URL	http://203.0.113.88:8080/payload.dll	Remote DLL fetched by certutil.exe (PID 5650)
File Artifact (Lure)	C:\Users\a.sharma\AppData\Local\Microsoft\Windows\INetCache\Content.Outlook\Q3_Tax_Adjustment.xlsm	Initial phishing attachment (SHA256: 9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08)
File Artifact (Payload)	C:\Users\a.sharma\AppData\Local\Temp\calc_helper.dll	Dropped DLL executed via rundll32.exe (SHA256: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855)
File Artifact (Persistence)	C:\Users\a.sharma\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\OneDriveUpdater.lnk	Malicious Startup shortcut (SHA256: a1b2c3d4e5f60718293a4b5c6d7e8f90123456789abcdef0123456789abcdef0)

Analyst Conclusion

The reconstructed telemetry presents a continuous sequence from the macro-enabled Office attachment through command-shell execution, LOLBin-based payload transfer and DLL execution, outbound C2 activity, discovery, and Startup-folder persistence. The incident is therefore documented in this lab as a CRITICAL-severity compromise requiring endpoint containment, artifact eradication, credential/session controls, network blocking, and preventive hardening.

Report Scope

This report is based on the simulated Splunk investigation data and the incident content supplied for Lab 3. The external IP address 203.0.113.88 is a documentation/test indicator in this lab scenario.